Section 2.1

Quadratic functions

First the definition of a polynomial.

Def

Let $n > 0$ $a_n, a_{n-1} \cdots a_1, a_0$ be real numbers Then the function:

$$f(x) = a_n x^n + a_{n-1} x^{n-1} + a_{n-2} x^{n-2} + \cdots + a_2 x^2 + a_1 x + a_0$$

is a polynomial of degree n

A Qudratic function is a polynomial of degree 2 ie $n = 2$

$$f(x) = a_2 x^2 + a_1 x + a_0$$

we often say that the General form for a Quadratic function is

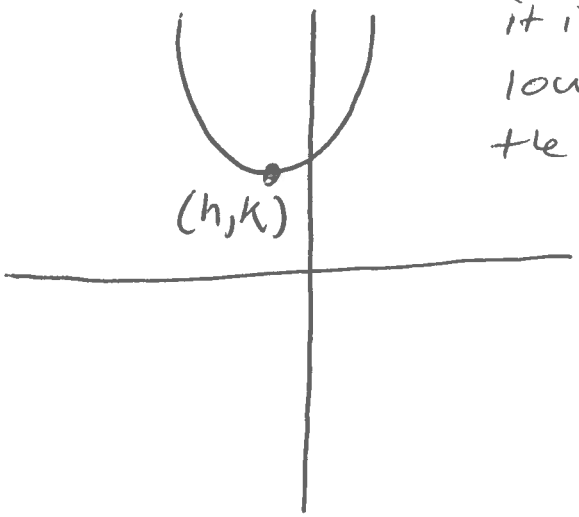
$f(x) = ax^2 + bx + c$ where a, b, c are real numbers

The standard form of a Quadratic function

$$f(x) = a(x-h)^2 + k$$

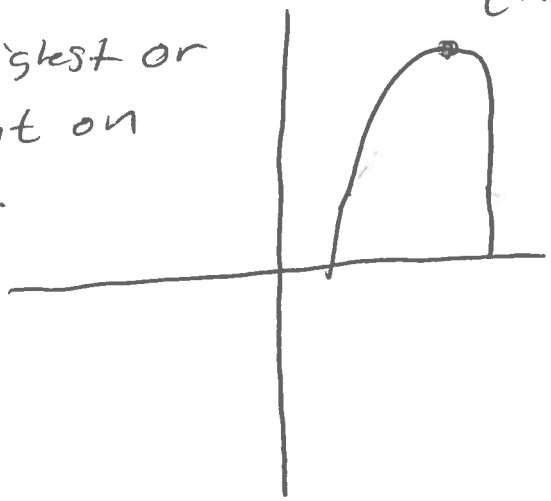
The point (h,k) is called the vertex it is the highest or lowest point on the graph.

$a > 0$



when $a > 0$ the parabola opens upward (h,k) is a minimum

$a < 0$



when $a > 0$ the parabola opens downward. (h,k) is a maximum

This form might look familiar in the context of function transformations

$$f(x) = (x+3)^2 + 2$$

↑ shift left 3

↑ shift up 2

Ex

for $f(x) = 3(x-2)^2 + 4$ what is the vertex

Sketch the graph.

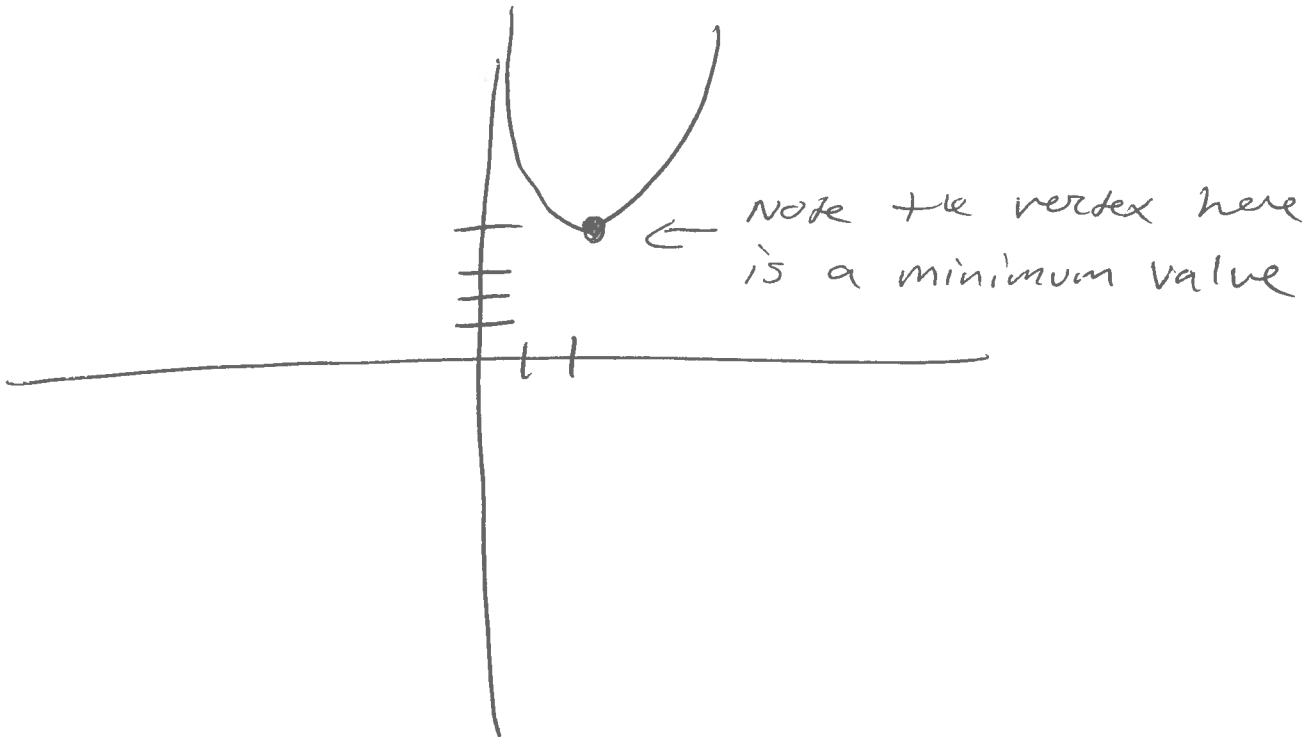
Soln

we match this to the general form

$$a(x-h)^2 + k$$
$$3(x-2)^2 + 4 \Rightarrow h=2 \quad k=4$$

so the vertex is at $(2,4)$ since $3>0$

The parabola opens upward



Ex

for $f(x) = -2(x+6)^2 - 3$ what is the vertex? Sketch the Graph.

Soln

Again we can match the forms

$$a(x-h)^2 + k$$
$$-2(x+6)^2 - 3$$

Note these signs are different we need them to be the same to read off (h, k) so we rewrite $f(x)$ as $-2(x-(-6))^2+(-3)$

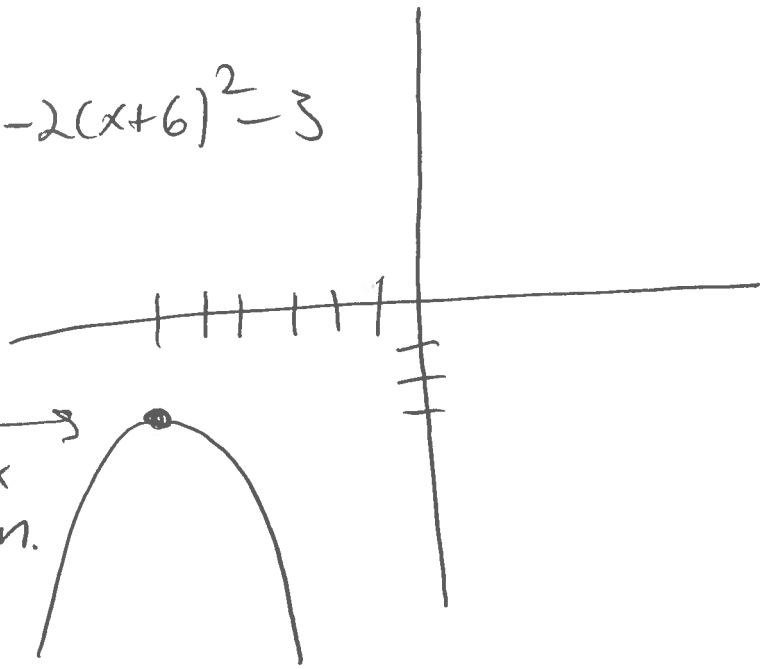
$$a(x-h)^2 + k$$
$$-2(x-(-6))^2 + (-3)$$

← since the signs are the same we can read off h, k

$h = -6$ $k = -3$ so the vertex is $(-6, -3)$

since $-2 < 0$ this opens downward

$f(x) = -2(x+6)^2 - 3$



Going from General form to Standard form.

Suppose we want to graph $f(x) = 2x^2 + 4x + 8$ it would be much easier if this were in Standard form. We can actually re-write $f(x)$ in Standard form by completing the Square. Lets Go through those

Steps

Completing the Square

Steps

Ex $f(x) = 2x^2 + 4x + 8$

① Factor out what ever you have to to get a coefficent of 1 infront of x^2

$2x^2 + 4x + 8 = 2(x^2 + 2x + 4)$

② Inside the parenthesis Take the coefficent of the x term divide it by 2 then Square it

$2(x^2 + \underset{\downarrow}{2}x + 4)$

$\frac{2}{2} = 1 \rightarrow 1^2 = 1$

③ add and subtract this number inside the parentlesides.

$+0$ so nothing has changed

$2(x^2 + 2x + 1 - 1 + 4$

④ Inside the parenthesis factor the perfect Square you have just created add the leftovers up

$2((x^2 + 2x + 1) - 1 + 4$

$2((x+1)^2 + 3)$

⑤ multply what you factored back through..

$2((x+1)^2 + 3)$

$= \boxed{2(x+1)^2 + 6}$

HW Example like #6 HW #2

a) complete the square for $3x^2+3x+4$

b) solve the resulting equation for the x intercepts

a) $3x^2+3x+4$

$3\left(x^2+x+\frac{4}{3}\right)$ ← factor out the 3 in front of x^2

$3(x^2+1x+4/3$

$\frac{1}{2} \rightarrow \left(\frac{1}{2}\right)^2 = \frac{1}{4}$ ← divide the coefficient of x by 2 and square it

$3\left(x^2+x+\frac{1}{4}-\frac{1}{4}+\frac{4}{3}\right)$ ← add and subtract it inside the parenthesis

$3\left(\left(x+\frac{1}{2}\right)^2+\frac{13}{12}\right)$ ←

$x^2+x+\frac{1}{4}=\left(x+\frac{1}{2}\right)^2$

$-\frac{1}{4}+\frac{4}{3}=\frac{-3}{12}+\frac{16}{12}=\frac{13}{12}$

$3\left(x+\frac{1}{2}\right)^2+3\frac{13}{12}$

$= \boxed{3\left(x+\frac{1}{2}\right)^2+\frac{13}{4}}$

b) Solve for x intercepts

Set = to 0

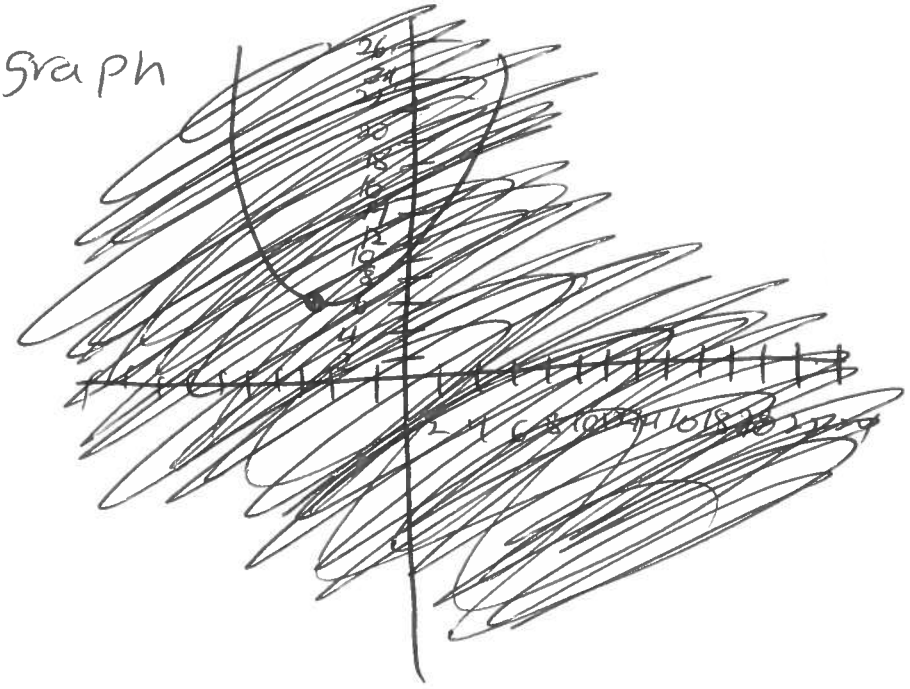
$3(x+1/2)^2 + \frac{13}{4} = 0$

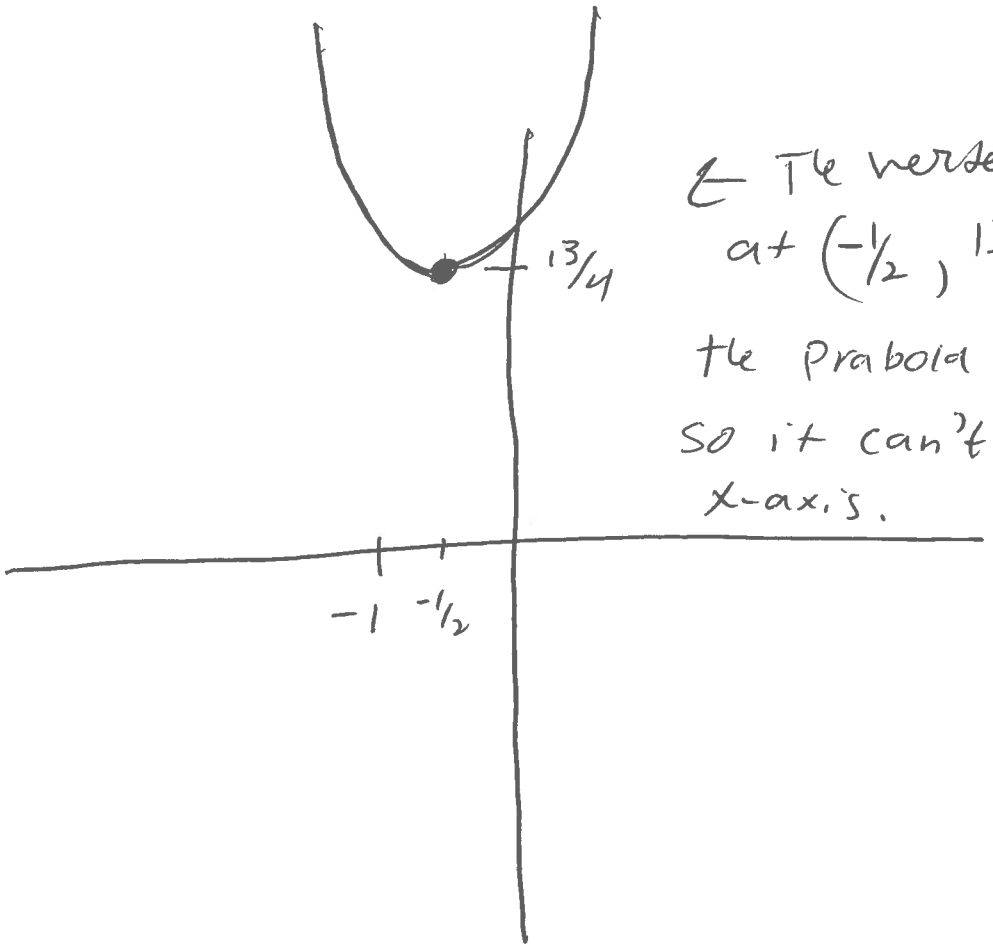
$\frac{3(x+1/2)^2}{3} = \frac{-\frac{13}{4}}{3}$

$(x+1/2)^2 = -\frac{13}{12}$

Uh oh... we would next take the square root of both sides but we would get imasinary numbers

So here we have no x intercepts this makes sense when we look at the graph



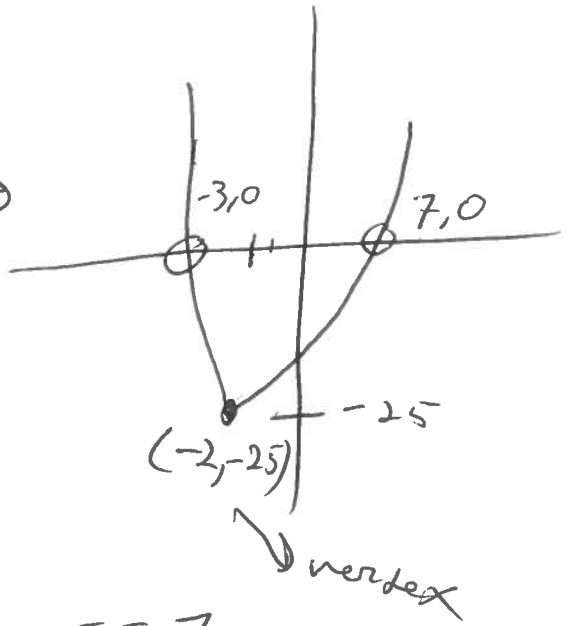


← The vertex is at $(-\frac{1}{2}, \frac{13}{4})$

the parabola opens up so it can't cross the x-axis.

HW Example like #7 HW 2

Solve $(x+2)^2 - 25 = 0$ ⟶



$+25 \quad +25$

$\sqrt{(x+2)^2} = \sqrt{25}$

$x+2 = \pm 5$

$-2 \quad -2$

$x = 2 \pm 5 \Rightarrow x = 2+5 = 7$

$x = 2-5 = -3$

Finding maximum and minium values.

For a parabola that opens upward the vertex is a minimum,

For a parabola that opens downward the vertex is a maximum.

Lets complete the square for the General form of a parabola

$$ax^2+bx+c$$

$$a\left(x^2+\left(\frac{b}{a}\right)x+\frac{c}{a}\right) \leftarrow \text{factor out } a$$

$$\hookrightarrow \frac{b/a}{2}=\frac{b}{2a} \rightarrow \left(\frac{b}{2a}\right)^2 \quad \text{Take coeff of } x \text{ divide by 2 square it}$$

$$a\left(x^2+\frac{b}{a}x+\left(\frac{b}{2a}\right)^2-\left(\frac{b}{2a}\right)^2+\frac{c}{a}\right) \leftarrow \text{add and subtract it}$$

$$=a\left(\left(x+\frac{b}{2a}\right)^2-\frac{b^2}{4a^2}+\frac{4ca}{4a^2}\right) \leftarrow \text{factor and add it up}$$

$$=a\left(\left(x+\frac{b}{2a}\right)^2+\frac{4ca-b^2}{4a^2}\right)$$

$$= a\left(x+\frac{b}{2a}\right)^2 + a\left(\frac{4ca-b^2}{4a^2}\right)$$

$$= a\left(x+\frac{b}{2a}\right)^2 + \frac{4ca-b^2}{4a}$$

Now we will rewrite this as

$$a\left(x-\left(\frac{-b}{2a}\right)\right)^2 + \frac{4ca-b^2}{4a}$$

and compare to: $a(x-h)^2 + k$

} matching signs to set h, k

So for ax^2+bx+c

$h = \frac{-b}{2a}$ $\quad k = \frac{4ca-b^2}{4a} \Rightarrow$ vertex is $\left(\frac{-b}{2a}, \frac{4ca-b^2}{4a}\right)$

Recall though that k is the y value of the function at the vertex

So $k = f(h) = f\left(\frac{-b}{2a}\right)$ ← slightly easier.

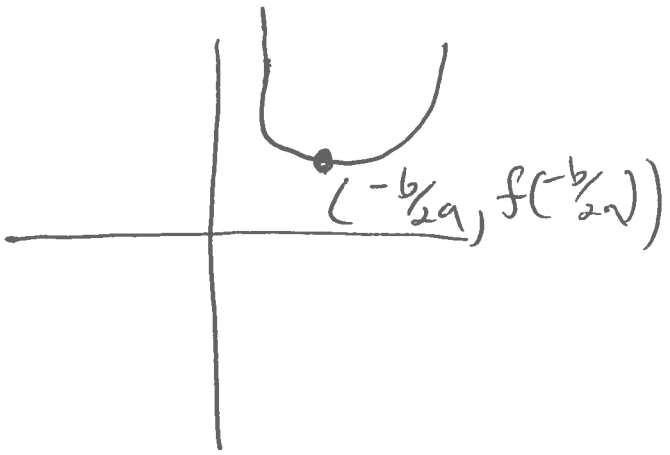
The y-value of the vertex is what we call the max or min so we can now say:

for $ax^2+bx+c=f(x)$

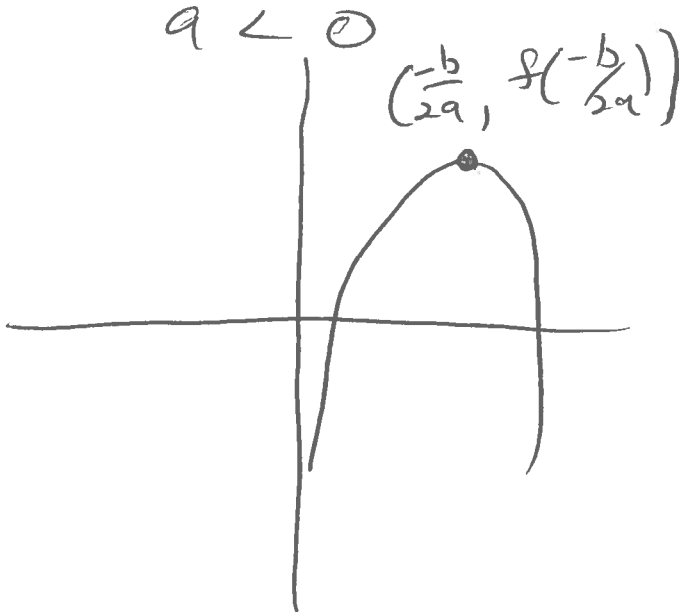
If $a>0$ f opens up and has vertex $(h,k)=\left(\frac{-b}{2a}, f\left(\frac{-b}{2a}\right)\right)$ and a minimum value at $f\left(\frac{-b}{2a}\right)$

If $a<0$ f opens downward and has vertex $(h,k)=\left(\frac{-b}{2a}, f\left(\frac{-b}{2a}\right)\right)$ with a maximum value at $f\left(\frac{-b}{2a}\right)$

$a>0$



$a<0$



Ex

Find the vertex of $f(x) = 4x^2 + 2x - 3$

what is its minimum value?

Soln

The vertex is at $\left(\frac{-b}{2a}, f\left(\frac{-b}{2a}\right)\right)$

$ax^2 + bx + c$

$4x^2 + 2x - 3$

$a = 4 \quad b = 2$

$$\frac{-b}{2a} = \frac{-2}{2(4)} = \frac{-2}{8} = \frac{-1}{4} = h$$

$$k = f\left(\frac{-b}{2a}\right) = f(-1/4) = 4(-1/4)^2 + 2(-1/4) - 3$$

$$= 4\frac{1}{16} - \frac{2}{4} - 3 = \frac{1}{4} - \frac{2}{4} - 3 = \frac{-1}{4} - 3 = \frac{-1}{4} - \frac{12}{4}$$

$$= \frac{-13}{4} = k \quad \text{so the vertex is } \left(-1/4, \frac{-13}{4}\right)$$

The minimum is $\boxed{f\left(\frac{-b}{2a}\right) = \frac{-13}{4}}$